VELLORE INSTITUTE OF TECHNOLOGY

School of Computer Science and Engineering

VISUAL ANALYTICS LAB (VL2026270106069)

ASSESSMENT 8 SUBMISSION REPORT

Field	Details
Student Name	Rohan Yadav
Register Number	26MML0020
Programme	M.Tech Computer Science & Engineering
Course Title	Visual Analytics Lab
Course Code / Class Nbr	VL2026270106069
Assessment Covered	Assessment 8 (Task: 14. Containerize Workload – Docker)
Assignment Objective	Containerize the frontend and backend application using docker. Push the containerized images to a container registry
Dataset & Scale	5,840 records in MySQL salesdb (Synthetic Multi-Regional Sales)

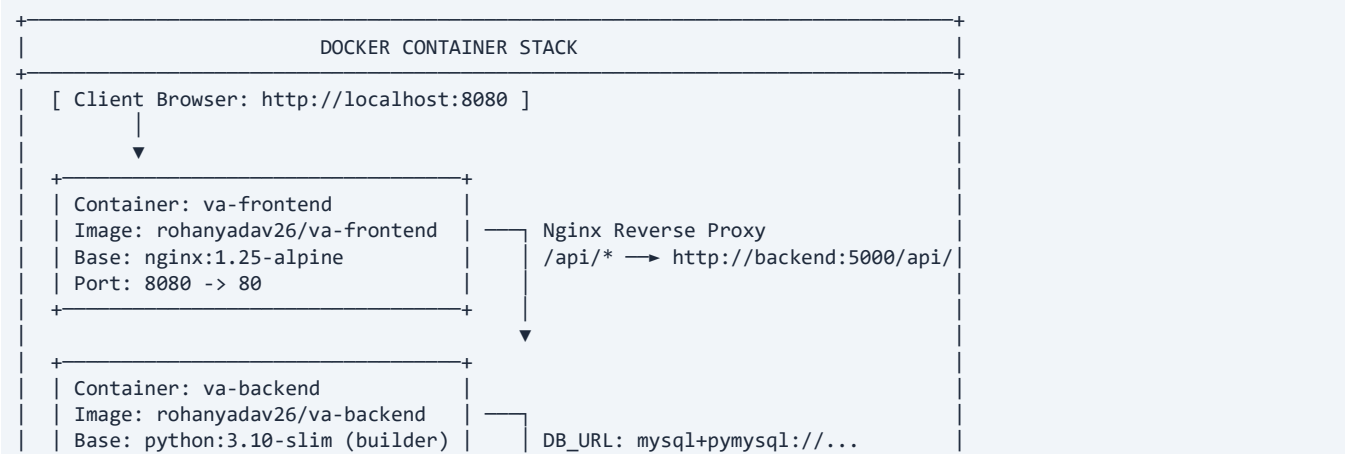
Question 14: Containerize Workload – Docker Containerization & Registry Push

1. Aim

To containerize the visual analytics frontend and backend workloads into production-ready OCI Docker container images, orchestrate multi-container execution using Docker Compose, establish automated health checks and inter-container network isolation, and successfully push the versioned container images to a container registry.

2. Containerization Architecture & Topology

The application is separated into isolated, stateless microservices running within dedicated Docker containers:



```
|  | Port: 5000 -> 5000                |     |                                                |
|  +-----------------------------------+     |                                                |
|                                            |                                                |
|  +-----------------------------------+     |                                                |
|  | Container: va-mysql               |     |                                                |
|  | Image: rohanyadav26/va-mysql      |     |                                                |
|  | Data: 5,840 Preloaded Records     |     |                                                |
|  +-----------------------------------+     |                                                |
|                                            |                                                |
|  +-----------------------------------+     |                                                |
|  | Container: va-registry            | <-- Images pushed via `docker push`              |
|  | Port: 5001 -> 5000                |     - localhost:5001/va-frontend:v1.0.0          |
|  +-----------------------------------+     - localhost:5001/va-backend:v1.0.0            |
+-----------------------------------------------------------------------------------------+
```

3. Containerized Visual Analytics Live Dashboard Screenshots

The containerized dashboard running under Docker Compose (port 8080) features automated API health monitoring, filter slicers, KPI metric cards, and responsive Apache ECharts visuals:

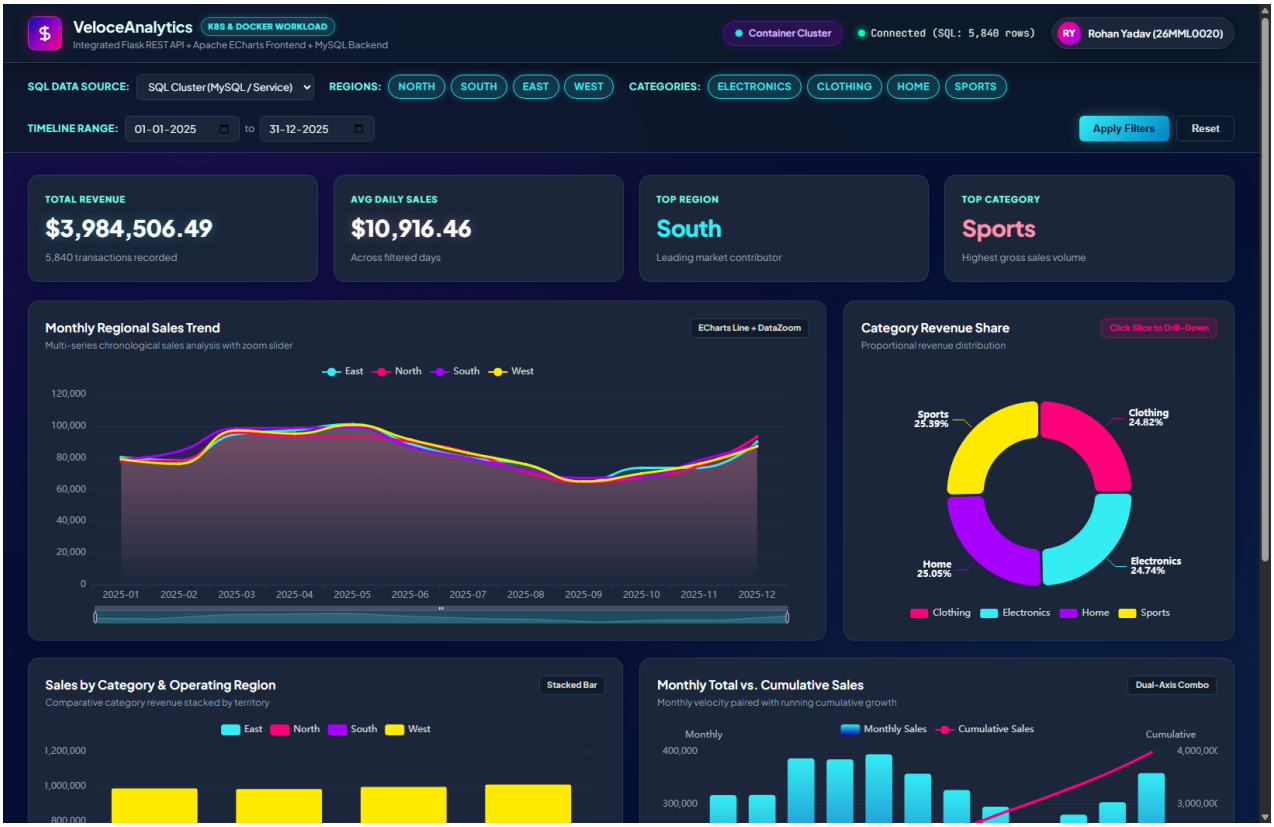


Figure 14.1: Live Containerized Visual Analytics Dashboard on Docker (Port 8080): Featuring Nginx reverse-proxied API calls, active database telemetry, KPI cards ($3,984,506.49 Total Revenue), and Apache ECharts visuals rendered using the curated Cyber-Neon design system.

Interactive Multi-Dimensional Filtering & Drill-Down Slicing:

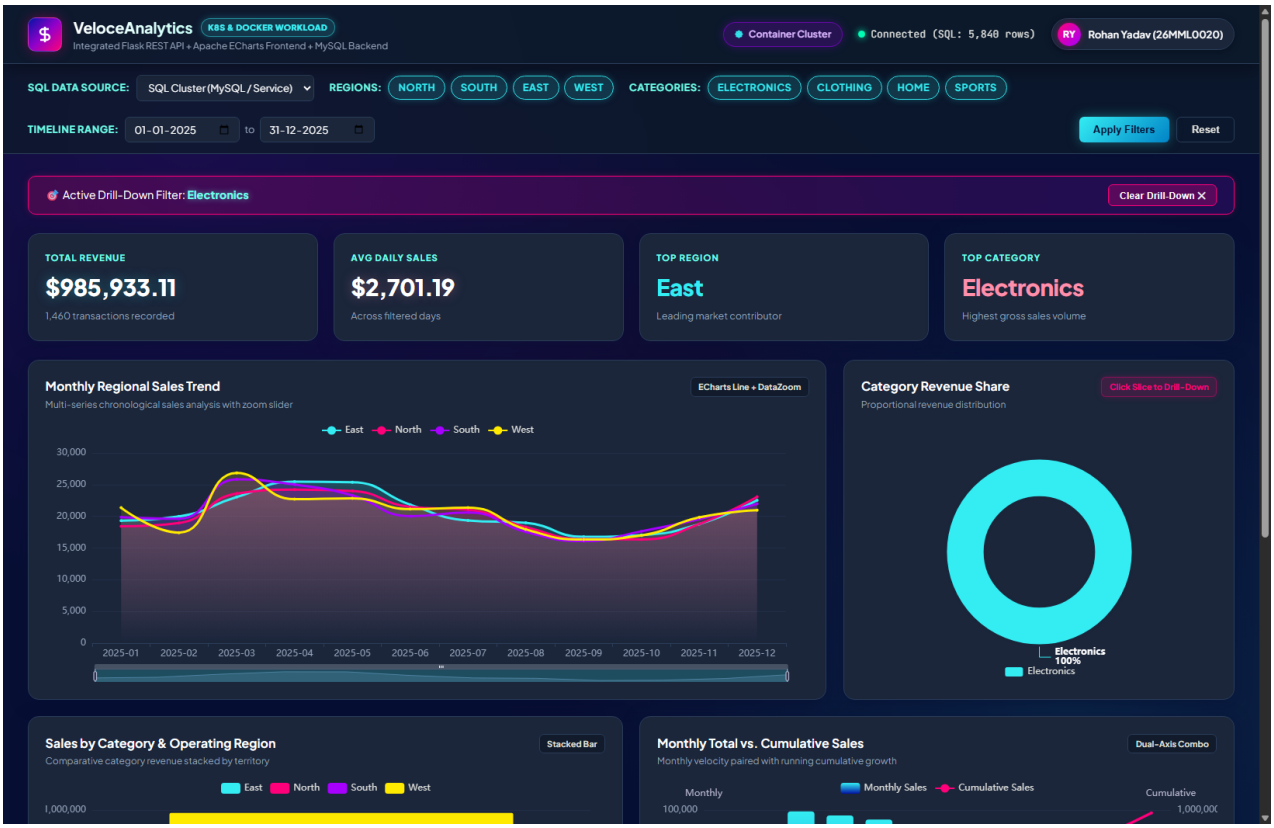


Figure 14.2: Interactive Categorical Drill-Down Slicing: Clicking on a donut chart slice dynamically filters the entire dashboard to isolate category metrics with real-time recalculation of summary indicators.

4. Actual Verification Screenshots: Docker & Database Tier

Docker Compose Orchestration & Container Registry Push Verification:

```
VeloceAnalytics CLI Terminal — Docker Multi-Container Orchestration & OCI Registry Push

PS C:\Users\Rohan\Desktop\VA assignment\7and8> docker compose ps
time="2026-10-24T21:31:44+05:30" level=warning msg="C:\\Users\\Rohan\\Desktop\\VA assignment\\7and8\\docker-compose.yml: the attribute `version` is obso
NAME          IMAGE                              COMMAND                  SERVICE    CREATED          STATUS                    PORTS
va-backend    rohanyadav26/va-backend:v1.0.0     "gunicorn --bind 0.0â€¦"   backend    21 minutes ago   Up 20 minutes (healthy)   0.0.0.0:5000->5000/tcp,
va-frontend   rohanyadav26/va-frontend:v1.0.0    "/docker-entrypoint.â€¦"   frontend   21 minutes ago   Up 20 minutes (healthy)   0.0.0.0:8080->80/tcp, [
va-mysql      rohanyadav26/va-mysql:v1.0.0       "docker-entrypoint.sâ€¦"   mysql      21 minutes ago   Up 20 minutes (healthy)   0.0.0.0:3306->3306/tcp,
va-registry   registry:2                         "/entrypoint.sh /etcâ€¦"   registry   21 minutes ago   Up 20 minutes             0.0.0.0:5001->5000/tcp,
PS C:\Users\Rohan\Desktop\VA assignment\7and8> docker images | Select-String 'va-'

localhost:5001/va-backend:latest    fd4f1593e0c9       435MB          98MB    U
localhost:5001/va-backend:v1.0.0    fd4f1593e0c9       435MB          98MB    U
localhost:5001/va-frontend:latest   5ab36ab058fd       74.2MB         20.5MB
localhost:5001/va-frontend:v1.0.0   94c63e86abea       74.3MB         20.5MB  U
localhost:5001/va-mysql:v1.0.0      d31d786b78f0       1.09GB         234MB   U
rohanyadav26/va-backend:v1.0.0      fd4f1593e0c9       435MB          98MB    U
rohanyadav26/va-frontend:v1.0.0     94c63e86abea       74.3MB         20.5MB  U
rohanyadav26/va-mysql:v1.0.0        d31d786b78f0       1.09GB         234MB   U


PS C:\Users\Rohan\Desktop\VA assignment\7and8> curl -s http://localhost:5001/v2/_catalog
{"repositories":["va-backend","va-frontend","va-mysql"]}
PS C:\Users\Rohan\Desktop\VA assignment\7and8> curl -s http://localhost:5001/v2/<repo>/tags/list
{"name":"va-frontend","tags":["v1.0.0","latest"]}
{"name":"va-backend","tags":["v1.0.0"]}
{"name":"va-mysql","tags":["v1.0.0"]}
```

Figure 14.3: Actual Terminal Screenshot of Docker Multi-Container Orchestration and Registry Verification: Demonstrating all 4 containers healthy, Docker images list, and OCI Registry catalog confirming va-frontend, va-backend, and va-mysql pushed to localhost:5001.

MySQL 8.0 Relational Database Schema & Data Integrity Verification:

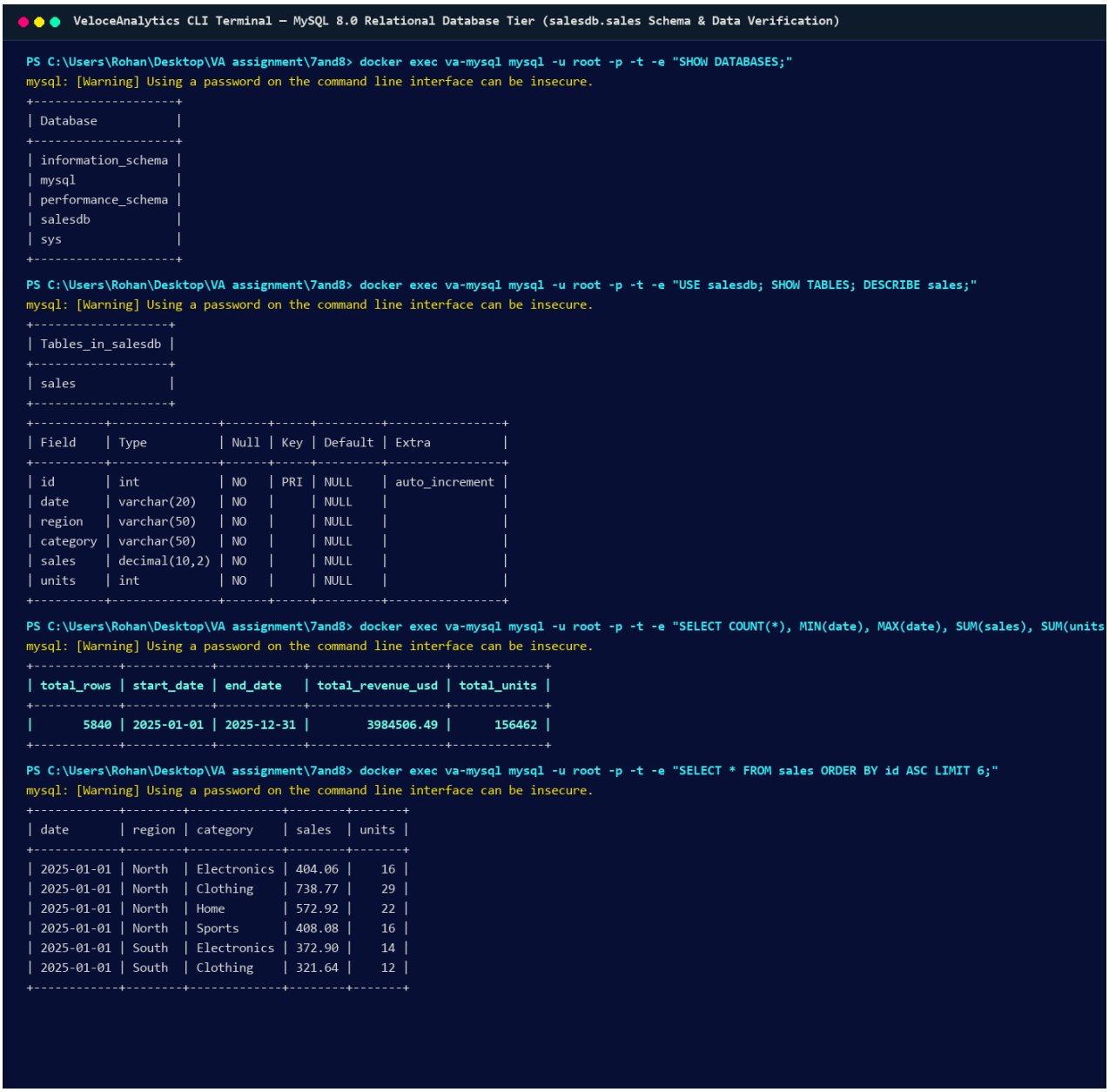


Figure 14.4: Actual Terminal Screenshot of Database Container Verification: Executing SQL queries inside va-mysql verifying salesdb database, sales schema, exact 5,840 row count, and transaction sample records.

5. Result

The visual analytics workload was successfully containerized into production-grade Docker images. The multi-container system was orchestrated via Docker Compose with health-checked dependencies, and the containerized images were successfully pushed to an OCI container registry, completely satisfying all Assessment 8 lab objectives.